

Algebraic Curves Lecture Notes

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1 Lecture 1 (October 1st 2026)

The majority of the class will be working over a general field k which is most often identified with $\mathbb{C}$. In fact, we will be working over characteristic 0 fields (ie there is no such $n \in \mathbb{Z}^+$ such that $n \cdot 1 = 0$). Sometimes the course will take place over $\mathbb{R}$, but this is primarily just for visualization.

Definition 1.1. An *affine algebraic set* is the set of points on which a finite collection of polynomials $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ vanishes. We write this as $V_k(f_1, \dots, f_m)$. Once again,

$$V_k(f_1, \dots, f_m) = \{\text{all } a \in k^n : f_i(a) = 0, \text{ for all } i\}$$

We moved onto examples after this, many of which were drawn to be visualized (which I might add here later).

Example 1.2. $V_{\mathbb{R}}(x^2 + y^2 - 1)$ is precisely the unit circle since it vanishes when $x^2 + y^2 = 1$.

Example 1.3. $(2, 3) \in \mathbb{R}^2$, the point, is defined by $V_{\mathbb{R}}(x - 2, y - 3) = \{(2, 3)\} \subseteq \mathbb{R}^2$ since these are clearly the roots of our two polynomials.

Example 1.4. $\mathbb{R}^2$ is the vanishing of 0's constant map: $V_{\mathbb{R}}(0)$.

Example 1.5. $\{a_1, \dots, a_n\} \subseteq \mathbb{C}$, points in $\mathbb{C}$ we can write this as the vanishing of $P(x)$ with these complex numbers as roots of said polynomial. We take $(x - a_1) \dots (x - a_n) = P(x)$ and $V_{\mathbb{C}}(P(x))$ is now our affine algebraic set for this set in $\mathbb{C}$

Example 1.6. The integers, $\mathbb{Z} \subseteq \mathbb{C}$ cannot be a affine algebraic set, because we need a finite number of finite degree polynomials which can only have finitely many roots.

Now, we move on to defining a more specific curve that the class will focus on.

Definition 1.7. An affine plane curve over k is a cut-out from a single non-constant polynomial in two variables.

$$C = V_k(P(x, y)) = \{(x, y) \in k^2 | P(x, y) = 0\}$$

This definition of a plane curve develops interesting questions for us to consider.

1. How can we describe the set of points of a given plane curve?
2. What can we say about the intersection of two plane curves? (Bezout's Theorem)
3. When do two polynomials describe the same plane curve? (Nullstellensatz)

Let's look at that last question carefully. We got some examples in class about this, and I will expand on it a little bit. In $\mathbb{R}$, we have $V_{\mathbb{R}}(x^2 + 1) = \emptyset$, and we have $V_{\mathbb{R}}(1) = \emptyset$ as well, but they have very little in common as polynomials. The interesting fact, however, is that this happens precisely because $\mathbb{R}$ is not algebraically closed, over $\mathbb{C}$ this is *less* of a problem. When we examine $V_{\mathbb{C}}(x^2 + 1) = \{-i, i\}$ and $V_{\mathbb{C}}(1) = \emptyset$ we see clearly that we have gotten rid of the problem.

We then define the degree of a polynomial which is the largest such $i + j$ you can find in the exponents of the polynomial

$$P(x, y) = \sum_{i,j} a_{ij} x^i y^j$$

The different kinds of plane curves we have are as follows

1. **Degree 1:** These are linear equations describable by $ax + by + c$ and the vanishing of that polynomial where a, b not both 0
2. **Degree 2:** These, like in high-school, are called conics and can be described generally by the vanishing of $P(x, y) = ax^2 + bxy + cy^2 + dx + ey + f$. A classic example is the unit circle, the vanishing of $x^2 + y^2 - 1$.
3. **Degree 2: Ellipse:** for example, the vanishing of $P(x, y) = x^2 + 4y^2 - 1$.
4. **Degree 2: Hyperbolas:** for example, the vanishing of $P(x, y) = x^2 - y^2 - 1$.
5. **Degree 2: Parabolas:** for example, the vanishing of $P(x, y) = x^2 - y$
6. **Degree 2: Degenerate Examples:** for example, the vanishing of xy or x^2 or y^2 which respectively, vanish on the union of the x and y axes, x axis, and y axis.

Now, we might be interested in certain solutions to intersections of lines with some of our plane curves. We look at an example from antiquity related to Pythagorean triples. We start by turning the Pythagorean theorem into a plane curve,

$$a^2 + b^2 = c^2 \implies \frac{a^2}{c^2} + \frac{b^2}{c^2} = 1$$

Hence, if we have pick $x = a/c$ and $y = b/c$ we can have $x^2 + y^2 = 1$ be our plane curve. So, we're really looking to rational points on the unit circle. Thus, pick $p_0 = (-1, 0)$ and we will examine a line going through the unit circle as a chord (secant line) and we will examine the solution to the second intersection which gives us a parameterization for the rest of the circle. This curve will look like $y = t(x + 1)$. Now, we have two equations

$$x^2 + y^2 = 1 \quad y = t(x + 1)$$

Substitution helps us get

$$x^2 + t^2(x + 1)^2 = 1 \implies (x + 1)(x - 1 + t^2(x + 1)) = 0 \implies x = \frac{1 - t^2}{1 + t^2}$$

Now, solving for y once again with substitution we get,

$$y = \frac{2t}{1 + t^2} \implies \text{Unit-Circle} = \left\{ (-1, 0) \cup \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right), t \in \mathbb{Q} \right\}$$

Since we want rational t we can pick $t = r/s$ and get,

$$\text{Unit-Circle} = \left\{ (-1, 0) \cup \left(\frac{s^2 - r^2}{s^2 + r^2}, \frac{2rs}{s^2 + r^2} \right) \right\}$$

Which means we know our a and b for checking pythagorean triples which is, (after multiplying over by c to clear denominators)

$$(s^2 - r^2, 2rs, s^2 + r^2), r, s \in \mathbb{Z}$$

But it is important to note that we will need to pick s, r with different parity or else we will have a triple that is all even. For example see if we pick 5, 7 for our s and r we get

$$(7^2 - 5^2, 2 \cdot 5 \cdot 7, 7^2 + 5^2) = (24, 70, 74)$$

which we can factor and see 12, 35, 37 is the triple that actually defines this one. Thus, what we need is not merely two co-prime integers, we want $\gcd(s, r) = 1$ and $s \bmod 2 \neq r \bmod 2$. Choosing these will give you infinitely many unique pairs since for any a which is even, there are infinitely many prime numbers p that are larger than a and hence coprime.

Exercise 1.8. *Why is this not true for degree 3?*

2 Lecture 2 (October 2nd 2026)

The majority of this class was reviewing some ideas from algebra and topology. We discuss ideals, the Hilbert Basis Theorem. Afterwards, we look at some algebra we can perform over varieties.

Definition 2.1. An *ideal* in a commutative ring R is $I \subseteq R$ which has two properties

1. Closure under addition: for any $a, b \in I$ we have $a + b \in I$
2. Closure under multiplication by elements of R such that for any $r \in R, a \in I$ we have $ra \in I$

Exercise 2.2. *The Ideal generated by a subset $X \subseteq R$ is the smallest ideal that contains all of X .*

Proof. Placeholder □

Example 2.3. The ideal generated by (1) the unit of R is equal to all of R .

Example 2.4. $R = \mathbb{Z}$ is a ring and we have ideals (n) for $n \in \mathbb{Z}$ because $\mathbb{Z}$ is a euclidean domain and hence a PID.

Example 2.5. in the ring $R = k[x, y]$ we have $(x - 3)$ and $(x - 3, y - 2)$ as ideals.

*** Aside:** The original hilbert basis theorem is not proven in our class, so we introduce and prove that here for completion's sake.

Definition 2.6. A *Noetherian* ring R is a ring that satisfies the ascending chain condition: for any sequence of Ideals (no left/right since we are only worried about commutative rings) $I_1 \subseteq I_2 \subseteq \dots \subseteq I_n \subseteq \dots$ there exists $m \in \mathbb{N}$ such that $I_m = I_{m+1} = I_{m+2} = \dots$. We say that the chain stabilizes.

Theorem 2.7 (Hilbert Basis Theorem). *If R is a noetherian ring then $R[X]$ is noetherian ring.*

Proof. Placeholder □

Theorem 2.8 (Our Class' Hilbert Basis Theorem). *For any field k , any ideal $I \subseteq k[x_1, \dots, x_n]$ has a finite generating set such that $I = (f_1, f_2, \dots, f_l)$*

* **Aside:** This version was also not proven in class so we will do it here:

Proof. Placeholder □

If $I \subseteq k[x_1, \dots, x_n]$ is an ideal then we write

$$V_k(I) = \{(a_1, \dots, a_n) \in k^n \mid f(a_1, \dots, a_n) = 0, \text{ for all } f \in I\}$$

Additionally by the Hilbert Basis Theorem since we know that I is finitely generated we have

$$I = (f_1, \dots, f_l)$$

We can also check that

$$I = \{g_1 f_1 + g_2 f_2 + \dots + g_l f_l : \text{for any } g_1, \dots, g_l \in k[x_1, \dots, x_n]\}$$

This helps motivate our next observation.

Remark 2.9. If $I = (f_1, \dots, f_l)$ then $V_k(I) = V_k(f_1, \dots, f_l)$

Proof. Placeholder □

Proposition 2.10. *Let V_1 and V_2 be affine algebraic sets in k^n then $V_1 \cap V_2$ is an affine algebraic set.*

Proof. Placeholder □

Exercise 2.11. *Let $V_1 = V_k(f_1, \dots, f_m)$ and $V_2 = V_k(g_1, \dots, g_l)$ be affine algebraic sets in k^n then $V_1 \cup V_2$ is an affine algebraic set where $V_1 \cup V_2 = V_k(g_j \cdot f_i \mid 1 \leq i \leq m, 1 \leq j \leq l)$*

Proof. Placeholder □

Quick check! Is the intersection of two affine plane curves also an affine plane curve? No. This is a key distinction between plane curve and affine algebraic set. The key distinction is that an affine algebraic set is defined by the vanishing of finitely many polynomial functions $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ but a plane curve is the vanishing of a single non-constant polynomial in two variables. Thus, we can look at

$$P(x, y) = x \quad Q(x, y) = y \implies V_k(P) \cap V_k(Q) = (0, 0)$$

Thus, for this to be a plane curve it has to be the vanishing of a single non-constant polynomial which is impossible.

Remark 2.12. The intersection over some indexing set $\bigcap_{\alpha} V(I_{\alpha})$ of arbitrarily many affine algebraic sets is still an affine algebraic set because $I_{\alpha} = (f_{\alpha,1}, \dots, f_{\alpha,i})$ so you can just look at the vanishing of all the generators

$$\bigcap_{\alpha} V(I_{\alpha}) = V\left(\sum_{\alpha} I_{\alpha}\right) \text{ since we have } \sum_{\alpha} I_{\alpha} = I\left(\bigcup_{\alpha} X_{\alpha}\right) \text{ is finitely generated by 2.8}$$

This remark lets us define the Zariski topology on affine space. Here we define the open sets on affine space as the complements of affine algebraic sets. Let's recall what we need to have for a topology,

1. $\emptyset$ and k^n are open sets.
2. finite intersections of open sets are open
3. arbitrary unions of open sets are open

Exercise 2.13. *Prove the Zariski Topology on affine space is indeed a topology.*

Proof. Placeholder □

* **Aside:** Here is a nasty contradicition proof we will need later, and is important for establishing that (0) defines the largest open set in the zariski topology later:

Exercise 2.14. For a given polynomial $f \in k[x_1, \dots, x_n]$, where $|k|$ is infinite, we can specialize the polynomial to a single variable polynomial $f^1 \in k[x]$ by evaluating it at some point $p = (p_1, \dots, p_{n-1}, x_n) \in \mathbb{A}^n$.

Proof. We proceed by induction on the number of elements in $k[x]$ this is already done. We now suppose the result holds in $n-1$ variables so that for n variables we have $f(x_1, \dots, x_n) \neq 0$. We write this polynomial as

$$f(x_1, \dots, x_n) = \sum_{j=0}^d g_j(x_1, \dots, x_{n-1})x_n^j$$

where not every $g_j = 0$, pick one and we can apply the $n-1$ case to that g_j and get

$$f(p_1, \dots, p_{n-2}, x_{n-1}, x_n) = \sum_{j=0}^d g_j(p_1, \dots, p_{n-2}, x_{n-1})x_n^j$$

Since each g_j is a polynomial of finite degree it can only have finitely many roots. Thus, we know, since k is infinite we can pick some p_{n-1} such that for at least one g_j , $g_j(p_1, \dots, p_{n-2}, p_{n-1}) \neq 0$. This then gives us

$$f(p_1, \dots, p_{n-2}, p_{n-1}, x_n) = \sum_{j=0}^d g_j(p_1, \dots, p_{n-2}, p_{n-1})x_n^j$$

Which is indeed a non-zero polynomial in one variable and we are done. $\square$

Exercise 2.15. When k is an infinite field we have that if an ideal $I \subseteq k[x_1, \dots, x_n]$ has the property that $V_k(I) = \mathbb{A}^n$ implies $I = (0)$.

Proof. We get as our hypothesis that $V_k(I) = \mathbb{A}^n$. Now suppose not for the sake of contradiction such that $I \neq (0)$. Pick some non-zero $f \in I$ which we know will vanish on x for $x \in \mathbb{A}^n$. We now specialize to f^1 using *Exercise 2.14*. We know that f^1 on $p = (p_1, \dots, x, \dots, p_n)$ will vanish over every $x = c \in k$, this gives us $f^1(c) = 0$ for any choice of c in $\mathbb{A}^n$ so we have some $f^1(c) = a_1c^d + a_2c^{d-1} + \dots + a_{d-1}c + a_d = 0$ for some finite d . But this is impossible since our one variable version of f can only have at most d many zeroes from our normal high-school algebra knowledge. But here we have infinitely many roots which is a contradiction. $\square$

This topology is weaker (she actually means coarser) than the typical topologies on $\mathbb{R}^n$ and $\mathbb{C}^n$. Indeed Zariski opens are open in the typical sense but open sets in the typical topologies are not necessarily open in Zariski topology.

* **Aside:**

Definition 2.16. A topology on X is called Hausdorff if for any two points $a, b \in X$ there exist U, V open sets such that $a \in U$ and $b \in V$ and $U \cap V = \emptyset$

Exercise 2.17. The Zariski Topology on $\mathbb{A}^n$, defined by $k[x_1, \dots, x_n]$ for k , where $|k|$ is infinite, is not Hausdorff.

Proof. Let, $a, b \in \mathbb{A}^n$ and let $U, V \subseteq \mathbb{A}^n$ be any two open sets around them so that we have $a \in U, b \in V$. We wish to show that these open sets will have an interection. We know that an

open set is defined by the complement of affine algebraic set in this topology. Thus, since we have assumed U, V non-empty we can write them as $U = \mathbb{A}^n - V_k(I)$ and $V = \mathbb{A}^n - V_k(J)$ where I, J ideals $\subseteq k[x_1, \dots, x_n]$. We now wish to show that these are not disjoint. Thus consider their intersection,

$$U \cap V = (\mathbb{A}^n - V_k(I)) \cap (\mathbb{A}^n - V_k(J)) = \mathbb{A}^n - (V_k(I) \cup V_k(J))$$

which we need to show is non-empty. But we know that the union of the vanishing is the set $V_k(I) \cup V_k(J) = V_k(g_j \cdot f_i | 1 \leq i \leq m, 1 \leq j \leq l)$ for $I = (f_1, \dots, f_m), J = (g_1, \dots, g_l)$ from *Exercise 2.11*. We know that the intersection is empty if the union of the vanishing is $\mathbb{A}^n$, but by *Exercise 2.15* we know this occurs when all $g_j \cdot f_i = 0$ for $1 \leq i \leq m, 1 \leq j \leq m$ which cannot occur since $g_i, f_i \in k[x_1, \dots, x_n]$ is an integral domain and not all g_i, f_i are 0 since $V_k(J)$ and $V_k(I)$ have non-empty complements in $\mathbb{A}^n$ they are both not the 0 ideal, (0) . Thus not all $g_j \cdot f_i$ are 0 and thus, the vanishing is not equal to the vanishing set over 0 and thus, the complement $U \cap V = \mathbb{A}^n - (V_k(I) \cup V_k(J)) \neq \emptyset$. And we are done. $\square$

*** Aside:** This is really bizarre and we need to be careful with exactly how we state it, and my lecturer is not being that careful. You specifically need $|k|$ to be infinite for this proof, and I think you can find counterexamples like $\mathbb{Z}/2$ which are not even pathological.

Theorem 2.18 (Fundamental Theorem of Algebra). $\mathbb{C}$ is algebraically closed.

*** Aside:**

Proof. Placeholder (may remove) $\square$

Proposition 2.19. if k is algebraically closed then any plane curve $C \subseteq k^2$ has infinitely many points over k ; if k is $\mathbb{C}$ then the curve is unbounded.

Lecture 3

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